FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. From the property of vector product, we notice that **C** must be perpendicular to the plane formed by vector **A** and **B**. Thus **C** must be perpendicular to **A** and to **B**. Thus the statements (1) and (2) are correct. Now **A** + **B** vector also must lie in the plane formed by vector **A** and **B**. Thus **C** must be perpendicular to **A** + **B** also the cross product (**A** × **B**) gives a vector **C** which cannot be perpendicular to itself. Thus the statement (4) is wrong.

2. D

3. A

Sol. The distance travelled by the rocket during burning interval (1 minute = 60 s) in which resultant acceleration is vertically upwards and is 10 m/s^2 will be

$h_1 = 0 \times 60 + (1/2) \times 10 \times 60^2 = 18000$ m ...(1)

And velocity acquired by it will be

$v = 0 + 10 \times 60 = 600$ m/s ...(2)

Now after 1 minute the rocket moves vertically up with initial velocity of 600 m/s and acceleration due to gravity oppose its motion.

So, it will go to a height h_2 till its velocity becomes zero such that

$0 = (600)^2 - 2gh_2$

$\Rightarrow$ $h_2 = 18000$ m [as $g = 10$m/s^2] ..(3)

So from eq. (1) and (3) the maximum height reached by the rocket from the ground.

$H = h_1 + h_2 = 18 + 18 = 36$ km (B) As after burning of fuel the initial velocity from Eq. (2) is 600 m/s and gravity opposes the motion of rocket, so from 1st equation of motion time taken by it to reach the maximum height (for which $v = 0$)

$0 = 600 - gt$, i.e.$t = 60$ s

after finishing of fuel, the rocket goes up for 60 sec i.e., 1 minute more.

Hence correct answer is (A)

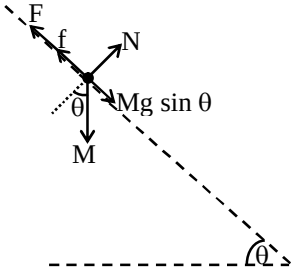
4. B

Sol. First data implies from

$\tan\theta = \mu_S$

that $\mu_S = \tan 30^\circ = \frac{1}{\sqrt{3}}$ .....(1)

For the second situation, the force, acting on the object are shown in fig. For equilibrium



$F + f_S = Mg\sin\theta$ .....(2)

$N = mg\cos\theta$

with $f_S = \mu_S Mg\cos\theta$ .....(3)

Using (1) and (3) in (2) with $\theta = 60^\circ$, we get

$F = Mg\sin\theta - \mu_S M\cos\theta\, g$

$= 60\sin 60^\circ - \frac{1}{\sqrt{3}}(60)\cos 60^\circ$

$= 30\sqrt{3} - \frac{30}{\sqrt{3}} = \frac{30\times 2}{\sqrt{3}}$ newton

or $F = \frac{30\times 2}{\sqrt{3}\times g}$ kg wt $= 2\sqrt{3}$ kg wt

5. C

Sol. The force acting on the particle = $\frac{mdv}{dt}$

Power of the force = $\left(\frac{mdv}{dt}\right) v = k$ (constant)

$m\frac{v^2}{2} = kt + c$ ...(1)

at $t = 0, v = u$ $\therefore$ $c = \frac{mu^2}{2}$

Now from (1),

$m\frac{v^2}{2} = kt + \frac{mu^2}{2}$

$\frac{1}{2}m(v^2 - u^2) = kt$ ...(2)

Again $\frac{mdv}{dt}v = k$

$m.v\frac{dv}{dx}v = k$

$\int_u^v mv^2 dv = \int_0^x k\,dx$

Intergrating, $\frac{1}{3}$m ($v^3 - u^3$) = kx ...(3)

From eqn (2) and (3),

t = $\frac{3}{2}\left(\frac{v^2-u^2}{v^3-u^3}\right)$ x

6. A

Sol. Distance $AD = \frac{4}{5}\times 100 = 80\,cm$

7. A

Sol. From ohm's law
Electric field α current density

8. D

Sol. $e = \int_{-\ell/2}^{\ell/2} B\omega r dr = 0$

9. A

10. B

11. B

Sol. When current is maximum $= \frac{di}{dt} = 0$

12. D

13. A

Sol. Saturation for A and B are different hence intensities are different. Stopping potentials for A and B are equal hence frequencies are equal.

14. B

Sol. Characteristic wavelengths shift of left if z is increased hence B is incorrect

15. D

16. C

17. B

Sol. $\rho_1 g h_1^2 = \rho_2 g h_2$

$\frac{h_1}{h_2} = \frac{\rho_2}{\rho_1}$

18. B

19. A

Sol. Process A $\rightarrow$ B is isothermal
Process B $\rightarrow$ C isochoric
Process C $\rightarrow$ A is isobaric

20. C

Sol. The process described is isochoric
Therefore molar heat capacity $= C_v = 2.5R$

21. A

22. B

Sol. Capacitors are in series therefore

$$\frac{C_1}{C_2}=\frac{V_2}{V_1}=\frac{2}{3}$$

23. A

Sol. Use $a=\frac{dV}{ds}$ and find $\frac{dV}{ds}$ using the slope of normal

24. B

25. A

Sol. $mu=3mv$

$V=u/3$

$$e=-\frac{2u/3-u/3}{-u}\Rightarrow e=1/3$$

26. D

Sol. Friction will be absent

$T=ma$

$mg-T=2ma$

Acceleration of $a=g/3$

Acceleration of block $=2g/3$

27. A

Sol. Since the scale is graduated at 10ºC,
1 cm of the scale at 10ºC = exactly 1 cm
∴ 1 cm of the scale at 30ºC
= exactly $(1 + 18 \times 10^{-6} \times 20)$cm
∴ 60 cm of the scale at 30ºC
= exactly $60.00\ (1 + 36 \times 10^{-5})$
= 60.02 cm

28. B

Sol. Tension at bottom end of rope = $T_1 = 2\times9.8$ N

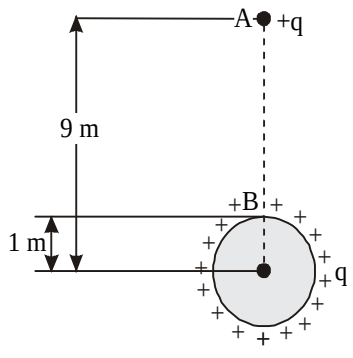
Θ weight of rope acts on gravity centre,
Therefore, tension at upper end of rope,
$T_2 = (6 + 2) \times 9.8 = 8 \times 9.8$ N
Thus, $T_2 = 4T_1$

if v_1 & v_2 are respective velocity at bottom & upper end, then $v_1 = \sqrt{\frac{T_1}{m}}$ & $v_2 = \sqrt{\frac{T_2}{m}}$

∴ $v_2 = 2v_1$ $(\Theta\ T_2 = 4T_1)$
Frequency n does not depend on medium,
therefore $v \propto \lambda$
if λ_1 & λ_2 are respective wavelength at bottom & upper end
$\lambda_2 = 2\lambda_1 = 2 \times 0.06 = 0.12$ m

29. A

Sol. Keeping in mind that here both electric and gravitational potential energy are changing and for external point a charged sphere behaves as whole of its charge were concentrated at its centre, applying conservation of energy between initial and final position, we have



$$\frac{1}{4\pi\varepsilon_0}\frac{qq}{9} + mg \times 9 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{1} + mg \times 1$$

or $q^2 = \dfrac{80\times10^{-3}\times9.8}{10^9}$

or $q = 28\ \mu C$

30. C

Sol. The fringe–width β is given by

$$\beta = \frac{\lambda D}{2d} \quad \text{and} \quad \beta' = \frac{\lambda D'}{2d}$$

where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

Now $\beta - \beta' = \dfrac{\lambda(D - D')}{2d}$

$\Rightarrow$ $\lambda = \dfrac{(\beta - \beta')2d}{D - D'}$

$$\lambda = \frac{(3\times10^{-5})(10^{-3})}{5\times10^{-2}}$$

$= 0.6 \times 10^{-6}$ m = 6000 Å

Chemistry

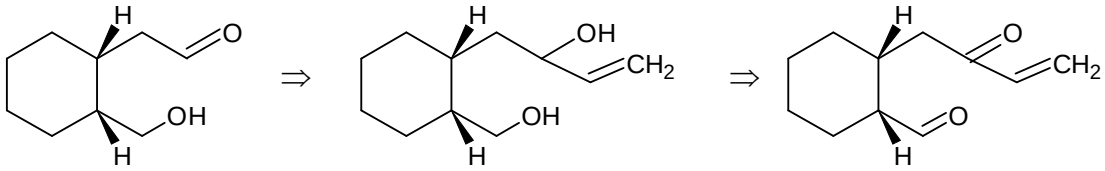
PART – II

SECTION – A

31. D
Sol. Theoretical

32. D
Sol. $S = C = C = C = S$

33. B
Sol.



34. B
Sol. Theoretical

35. C
Sol. Theoretical

36. A
Sol. $C(s) + 2Cl_2(g) \longrightarrow CCl_4(\ell) \qquad \Delta H = -135.5\,KJ/mole$

$CCl_4(\ell) \longrightarrow CCl_4(g) \qquad \Delta H = 30.5\,KJ/mole$

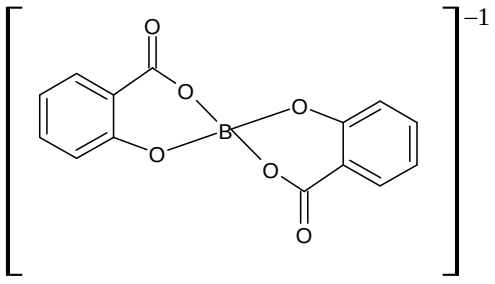
$C(s) + 2Cl_2(g) \longrightarrow CCl_4(g) \qquad \Delta H = -105\,KJ/mole$

$715 + 2\times 242 - 4\times x = -105$

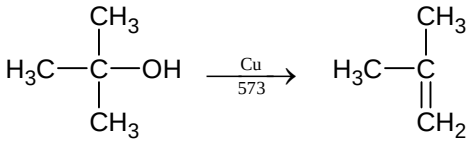
$x = 326\,KJ/mole \qquad i.e.\, x = B.E.of\ C-Cl\ in\ CCl_4(g)$

37. D
Sol. The total energy of the system takes into account the kinetic energies of all the sub-atomic particles, attractive potential between the electrons and nuclei also **include** repulsive potential among the electrons and nuclei individually.

38. B
Sol. the product is



39. D
Sol.



i.e. dehydration

40. B

Sol. Polymer is prepared by condensation reaction and cross link polymer.

41. D
Sol. Theoretical

42. A
Sol. Theoretical

43. D
Sol. VSEPR theory

44. A
Sol. $Ni(OH)_2(s) \rightleftharpoons Ni^{2+} + 2OH^-$

$s \qquad 2s+0.1$

$s(2s+0.1)^2 = 2\times10^{-15} \qquad 2s+0.1 \approx 0.1$

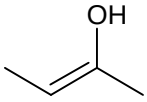
$s = 2\times10^{-13}$

45. B
Sol. The process is non- spontaneous at given condition so $\Delta G > 0$

46. D
Sol. Malachite is $CuCO_3.Cu(OH)_2$

47. C
Sol. Theoretical

48. B
Sol. One M^{3+} ion can replace three Na^{+1} ion so the cationic vacancies produced are equal to twice the number of M^{3+} ions.

49. C
Sol. 

this enole form is more stable

50. A
Sol. Reaction goes by S_N^1 mechanism so effect of increase in concentration of CH_3COO^- does not increase rate of reaction.

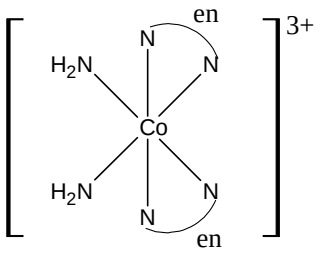
51. C
Sol. Theoretical

52. B
Sol. Theoretical

53. D
Sol. As the oxidation state of metal decrease the extant of back bonding by CO increases.

54. B

Sol.



55. B
Sol. Because Pb^{2+} ion will interfere due to formation of $PbSO_4$.

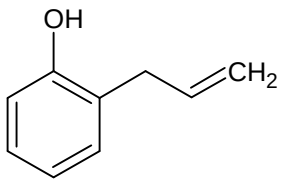
56. C
Sol. Bismuth(III) compounds hydrolysed to give oxymetalic $[BiO]^{+1}$ ion.

57. D
Sol. Theoretical

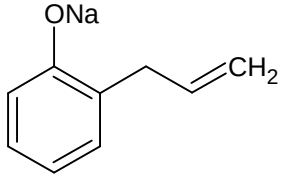
58. A
Sol. Prussian blue is a deep blue pigment is $Fe_4\left[Fe(CN)_6\right]_3$.

59. C
Sol. Theoretical

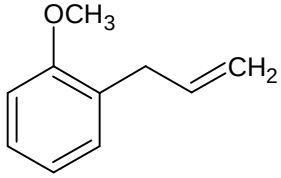
60. B
Sol.



A



B



C

Mathematics

PART – III

SECTION – A

61. D

Sol. $f(x)=x^4+2x^3-x^2+1$ is polynomial of even degree hence its range can't be R.

Hence not surjective.

$f(x)=x^3+x+1,\ f'(x)=3x^2+1 \Rightarrow$ monotonic hence bijective.

$f(x)=\sqrt{1+x^2}$, neither surjective not injective.

$f(x)=x^3+2x^2-x+1,\ f'(x)=3x^2+4x-1 \Rightarrow D = 16 + 12 > 0.$

Hence $f(x)$ is non-monotonic cubic polynomial hence surjective but not injective.

62. B

Sol. For $0 < x < \frac{\pi}{2}$ or $\frac{\pi}{2} < x < \pi$, $0 < \sin x < 1$.

$\therefore$ for $n > 1$, $\sin x > \sin^n x$

$\therefore \quad f(x)=\left[\frac{3(\sin x-\sin^n x)}{\sin x-\sin^n x}\right]=3,\ x\neq\frac{\pi}{2}$ and $f(x) = 3,\ x=\frac{\pi}{2}$

Thus in $(0, \pi)$, $f(x) = 3$.

Hence $f(x)$ is continuous and differentiable at $x=\frac{\pi}{2}$.

63. C

Sol. If $x_0 = K \in I$, then $\lim\limits_{x\to K+} f(x)=\lim\limits_{x\to K} K+\sqrt{x-K}=K$

and $\lim\limits_{x\to K-} f(x)=\lim\limits_{x\to K-}(K-1)+\sqrt{x-(K-1)}$

$=\lim\limits_{x\to K}(K-1)+\sqrt{x-K+1}=K-1+1=K$

Hence f is continuous at every $x_0 = K \in I$. If $x_0 \in R \sim I$, then $[x]$ is continuous at x_0 which is turn gives that f is continuous at every $x_0 \in R \sim I$

64. A

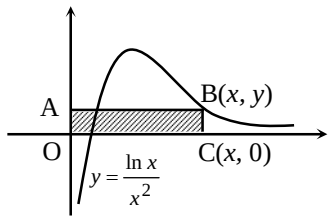
Sol. Area = $A = xy$

$A=\frac{\ln x}{x} \Rightarrow \frac{dA}{dx}=\frac{1-\ln x}{x^2}$

$\frac{dA}{dx}=0 \Rightarrow x = e$

$\frac{dA}{dx}<0,\ \forall x>e$ and $\frac{dA}{dx}>0,\ \forall x<e$.

Hence $x = e$ is point of maxima. $A_{\max}=e.\frac{1}{e^2}=\frac{1}{e}$



65. B

Sol. $f(x)=\int\frac{x^2(\sqrt{1+x^2}-1)}{(1+x^2)(1+x^2-1)}dx=\int\frac{\sqrt{1+x^2}-1}{1+x^2}dx=\int\frac{dx}{\sqrt{1+x^2}}-\int\frac{dx}{1+x^2}$

$=\log(x+\sqrt{1+x^2})-\tan^{-1}x+k$

$\therefore\ f(0) = \log 1 - \tan^{-1} 0 + k = k = 0$

$\therefore\ f(x) = \log(x + \sqrt{1+x^2}) - \tan^{-1} x$

$\therefore\ f(1) = \log(1+\sqrt{2}) - \dfrac{\pi}{4}$

66. B

Sol. Let $I = f\left(\dfrac{1}{2}\right) - f\left(\dfrac{1}{3}\right) = \int_0^{\pi/4} \log_e \left(\dfrac{1+\frac{1}{2}\tan z}{1+\frac{1}{3}\tan z}\right) dz = \int_0^{\pi/4} \log_e \left(\dfrac{3}{2}.\dfrac{2+\tan z}{3+\tan z}\right) dz$

$$= \int_0^{\pi/8} \left(\log_e \left(\frac{3}{2}.\frac{2+\tan z}{3+\tan z}\right) + \log_e \left(\frac{3}{2}.\frac{2+\tan\left(\frac{\pi}{4}-z\right)}{3+\tan\left(\frac{\pi}{4}-z\right)}\right)\right) dz = \frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$$

$\Rightarrow\ I = \dfrac{\pi}{8}\log_e\left(\dfrac{9}{8}\right)$

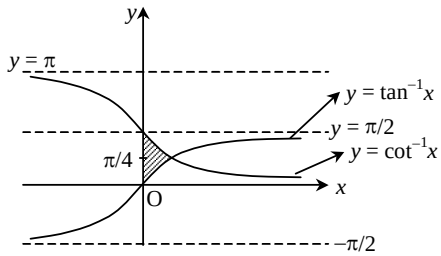
67. D

Sol. Required area,

$$\Delta = \int_0^{\pi/4} \tan y\, dy + \int_{\pi/4}^{\pi/2} \cot y\, dy$$

$$= \ln \sec y \Big|_0^{\pi/4} + \ln \sin y \Big|_{5\pi/4}^{\pi/2}$$

$$= \ln \sqrt{2} - \ln \frac{1}{\sqrt{2}} = \ln 2\,.$$



68. C

Sol. Put $\tan y = z\ \ \therefore\ \sec^2 y \dfrac{dy}{dx} = \dfrac{dz}{dx}\ \ \therefore\ \dfrac{dz}{dx} + xz = x^3$

which is a linear differential equation, on solving we get

$\tan y = x^2 - 2 + ce^{-x^2/2}$

69. B

Sol. Numerator = $\cos 6x + \cos 4x + 5(\cos 4x + \cos 2x) + 10(\cos 2x + 1)$

= $2\cos x\{\cos 5x + 5\cos 3x + 10\cos x\}$

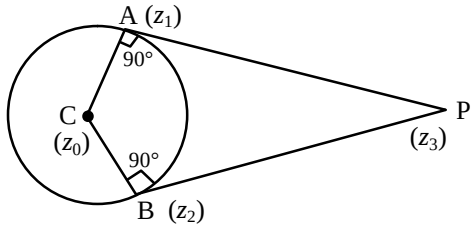
$\therefore\ y = \dfrac{\text{Nr}}{\text{Dr}} = 2\cos x,\ \dfrac{dy}{dx} = -2\sin x$

70. C

Sol. Let z_1, z_2 are represented by *A, B* whereas

z_0 is represented by *C*.

Let P represent z_3



$$\frac{z_3 - z_1}{z_0 - z_1} = \frac{PA}{AC} e^{\frac{i\pi}{2}} ; \frac{z_0 - z_2}{z_3 - z_2} = \frac{BC}{PB} e^{\frac{i\pi}{2}}$$

$$\frac{z_3 - z_1}{z_0 - z_1} \cdot \frac{z_0 - z_2}{z_3 - z_2}$$

$$= \frac{PA}{\text{radius}} \cdot \frac{\text{radius}}{PB} e^{i\pi} = -1 \ (\because PA = PB)$$

71. B

Sol. Let A_1 be the property that HIN appears

Let A_2 be the property that DUS appears

Let A_3 be the property that TAN appears

$n(A_1) = 7! = 5040$

$n(A_2) = \frac{7!}{2!} = 2520$

$n(A_3) = 7! = 5040$

$n(A_1 \cap A_2) = n(A_2 \cap A_3) = n(A_3 \cap A_1) = 5! = 120$

$n(A_1 \cap A_2 \cap A_3) = 3! = 6$

$n(A_1 \cup A_2 \cup A_3) = \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + n(A_1 \cap A_2 \cap A_3) = 12246$

Total number of ways to arrange the letters of the word HINDUSTAN $= \frac{9!}{2!} = 181440$

$\therefore$ Number of ways in which neither the pattern HIN or DUS and TAN appears
= 181440 – 12246 = 169194

72. A

Sol. Total number of matrices = 3^9
Number of symmetric matrices = 3^6
Number of skew symmetric matrices = 3^3
The zero matrix is both symmetric and skew-symmetric the required probability is

$$= \frac{3^9 - 3^6 - 3^3 + 1}{3^9}$$

73. A

Sol. LHS achieves its maximum value at $x = 1$ (in the domain of the equation i.e., $x > 0$)
Maximum value of LHS =1 at $x = 1$
RHS achieves it minimum value at $x = 1$ (in the domain of equation i.e., $x > 0$)
Minimum value of RHS = 1 at $x = 1$
Hence only possible value of x satisfying the given equation is 1.

74. D

Sol. $(1 - x + x^2)^n = a_0 - a_1 x + a_2 x^2 - .... + a_{2n} x^{2n}$

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + ... + {}^nC_n x^n$

On multiplying and collecting coefficient of x^{3r}, we get $\sum_{k=0}^{3r} (-1)^k a_k \, {}^nC_{3r-k} = {}^nC_r$

75. A

Sol. We can write Δ as

$$\Delta = \begin{vmatrix} 0 & \sin x - \cos x & \log \tan x \\ -(\sin x - \cos x) & 0 & -\log \tan x \\ -\log \tan x & \log \tan x & 0 \end{vmatrix}$$

Which is determinant of a skew symmetric matrix of odd order, hence value will be zero

76. B

Sol. $A\,(\text{Adj } A) = |A| I_3$

$K = |A| = 2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x)$

$= 2(1 - 3\sin^2 x.\cos^2 x) - 3(1 - 2\sin^2 x \cos^2 x) = -1$

77. C

Sol. By the given condition $(1-ab)m^2 - (a^2 + b^2)m - (1+ab) = 0$

$\Rightarrow m(a^2 + b^2) + (m^2 + 1)ab = m^2 - 1$ ...(i)

Now H_1 = First H.M. between *a* and *b* = $\frac{(m+1)ab}{a+mb}$ and $H_m = \frac{(m+1)ab}{b+ma}$

$\therefore H_m - H_1 = (m+1)ab\left[\frac{1}{b+ma} - \frac{1}{a+mb}\right] = (m+1)ab\frac{[(m-1)(b-a)]}{(b+ma)(a+mb)}$

=

$\frac{(m^2-1)ab(b-a)}{m(a^2+b^2)+(m^2+1)ab} = \frac{(m^2-1)ab(b-a)}{m^2-1}$ [From (i)]

= $ab(b-a)$.

78. B

Sol. We have, $a_n = S_n - S_{n-1}$ ...(i)

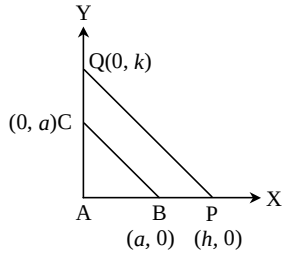
and $a_{n-1} = S_{n-1} - S_{n-2}$ ...(ii)

$\therefore d = a_n - a_{n-1} = (S_n - S_{n-1}) - (S_{n-1} - S_{n-2})$ [From (i) and (ii)]

$= S_n - 2S_{n-1} + S_{n-2}$

79. C

Sol. We take *A* as the origin and *AB* and *AC* as *x*-axis and *y*-axis respectively. Let *AP* = *h*, *AQ* = *k* Equation of the line *PQ* is



$\frac{x}{h} + \frac{y}{k} = 1$ ...(i)

Given, $BP \cdot CQ = AB^2$

$\Rightarrow (h-a)(k-a) = a^2$

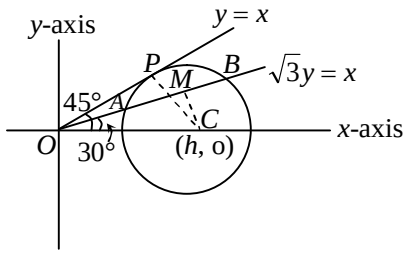
$\Rightarrow hk - ak - ah + a^2 = a^2$ or $ak + ha = hk$

Or $\frac{a}{h} + \frac{a}{k} = 1$ ....(ii)

From (ii), it follows that line (i) i.e. PQ always passes through a fixed point (*a*, *a*).

80. A

Sol. $OP = CP = \frac{h}{\sqrt{2}} = AC$



$CM = h\sin 30° = \frac{h}{2}$

From Δ *ACM*

$AC^2 = AM^2 + CM^2$

$\frac{h^2}{2} = 1 + \frac{h^2}{4} \Rightarrow \frac{h^2}{4} = 1 \quad \Rightarrow h = 2$

radius $= \sqrt{2}$

$(x-2)^2 + y^2 = 2 ; x^2 + y^2 - 4x + 2 = 0$

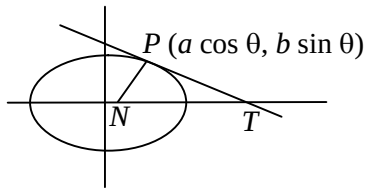
81. B

Sol. Mid-point of (1, 2) and (3, 6) is (2, 4), line passing through (1, 6) and (2, 4) is parallel to axis of the parabola (property). So, slope of its axis is –2 or slope of its directrix is 1/2, also in this case tangents are perpendicular. So, (1, 6) must lie on directrix,

$\therefore\ y - 6 = \frac{1}{2}(x-1) \quad \Rightarrow \quad x - 2y + 11 = 0$

82. A

Sol. $e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{3}{5}$



$T \equiv \left(\frac{a}{\cos\theta}, 0\right),\ N \equiv \left(\frac{a^2 - b^2}{a\sec\theta}, 0\right)$

On solving we get, $\cos\theta = \frac{1}{2},\ \sin\theta = \frac{\sqrt{3}}{2}$

$\therefore$ Area of $\Delta PSS' = 6\sqrt{3}$ sq. units

83. C

Sol. $b^2 = d_1 d_2$, where d_1 and d_2 are perpendicular distances of any tangent from focii of the hyperbola $b^2 = 4 \times 2 = 8 \Rightarrow b = 2\sqrt{2}$

$2ae = \sqrt{72} = 6\sqrt{2}$

So $a^2 = 10$

Centre is (5, 1)

Equation of conjugate axis $x - y = 4$ and equation of transverse *axis* is $x + y = 6$

$\therefore$ Equation of hyperbola is $\frac{(x-y-4)^2}{20} - \frac{(x+y-6)^2}{16} = 1$

84. B

Sol. The expression = $\sin^2 A + 2\sin A\cos B\sin C + \cos^2 B$

$= \sin^2 A + 1 - \sin^2 B + 2\sin A\cos B\sin C$

$= \sin(A+B)\sin(A-B) + 1 + 2\sin A\cos B\sin C$

$= -\sin C\sin(A-B) + 2\sin A\cos B\sin C + 1$

$= \sin C[2\sin A\cos B - \sin(A-B)] + 1$

$= \sin C\sin(A+B) + 1$

$= \sin C(-\sin C)+1=\cos^2 C$

85. B

Sol. $\frac{\sin \pi/12}{\sin x}+\frac{\cos \pi/12}{\cos x}=2 \Rightarrow \sin\frac{\pi}{12}\cos x+\cos\frac{\pi}{12}\sin x=2\sin x\cos x$

$\sin\left(\frac{\pi}{12}+x\right)=\sin 2x$

$\frac{\pi}{12}+x=2x$ and $\frac{\pi}{12}+x=\pi-2x$

$\Rightarrow x=\frac{\pi}{12}$ and $x=\frac{11\pi}{36}$

86. C

Sol. Adding the given two

$\sin(B+C)=1 \Rightarrow B+C=90°$

Subtracting, $\sin(B-C)=\frac{\sqrt{3}-2}{\sqrt{3}}\neq 0$

$\therefore B\neq C$

$\therefore$ Triangle is right-angled.

87. A

Sol. Put $\tan^{-1}x=\theta$ $\tan^{-1}x=\theta\Rightarrow\theta\in(-\pi/2,-\pi/4)\Rightarrow 2\theta\in(-\pi,-\pi/2)$

$\therefore$ equation becomes π-$\cos^{-1}(\cos 2\theta)+\sin^{-1}(\sin 2\theta)+\tan^{-1}(\tan 2\theta)=\pi/3$

On solving we get $\theta=-\pi/3\Rightarrow x=-\sqrt{3}$

88. B

Sol. The equation of a plane parallel to the plane $\vec{r}\,.(4\hat{i}-12\hat{j}-3\hat{k})-7=0$ is

$\vec{r}\,.(4\hat{i}-12\hat{j}-3\hat{k})+\lambda=0$. This passes through $2\hat{i}-\hat{j}-4\hat{k}$. Therefore

$(2\hat{i}-\hat{j}-4\hat{k})\,.(4\hat{i}-12\hat{j}-3\hat{k})+\lambda=0$

$\Rightarrow 8+12+12+\lambda=0 \Rightarrow \lambda=-32$

So, the required plane is $\vec{r}\,.(4\hat{i}-12\hat{j}-3\hat{k})-32=0$

89. A

Sol. $p\rightarrow q$ must be true & q→r must be false

$\therefore$ q must be true, r must be false but p can be true or false.

90. C

Sol. Deviation does not changes by changing a data